

Lecture 01 - Introduction

This lecture is the orientation layer for the whole course. It explains why interactive computer graphics is a pipeline problem: a scene must be represented, processed under time constraints, sampled into an image, and displayed fast enough for interaction.

Source chunk	course_text_parts/03_lectures/01_introduction.txt
Extracted slide pages	51
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Per-slide commentary

Each extracted slide page gets its own note. Use this section as the spoken commentary that the terse slide deck is missing.

Page 1	Untitled slide
Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Commentary	This slide is about Untitled slide. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Untitled slide' into a causal sentence instead of repeating the slide title?

Page 2	Course Learning Outcomes
Source cue	- Learn to program computer graphics applications using OpenGL / - Learn to realize efficient computer graphics applications on the GPU / - Implement your own computer graphics project / [Image Credits: Wikipedia User Mimigu] [Crassin et al., PG2011]
Commentary	This slide is about Course Learning Outcomes. Treat it as relevance information. It tells you how the lecture, exercises, programming tasks, and exam preparation connect. The visible cue is: - Learn to program computer graphics applications using OpenGL / - Learn to realize efficient computer graphics applications on the GPU / - Implement your own computer graphics project / [Image Credits: Wikipedia User Mimigu] [Crassin et al., PG2011]
Why it matters	Course logistics often reveal which topics are practiced, assessed, or expected in code.

Check yourself	Can you connect 'Course Learning Outcomes' to an exercise, project task, or exam-preparation action?
Page 3	Real-Time Rendering Contest Results 1/3
Source cue	Students: Frank Richter, Manuel Herbert Güntzel
Commentary	This slide is about Real-Time Rendering Contest Results 1/3. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: Students: Frank Richter, Manuel Herbert Güntzel
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Real-Time Rendering Contest Results 1/3' into a causal sentence instead of repeating the slide title?
Page 4	Real-Time Rendering Contest Results 2/3
Source cue	Students: Andreas Rottach, Matthias Englert
Commentary	This slide is about Real-Time Rendering Contest Results 2/3. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: Students: Andreas Rottach, Matthias Englert
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Real-Time Rendering Contest Results 2/3' into a causal sentence instead of repeating the slide title?
Page 5	Real-Time Rendering Contest Results 3/3
Source cue	Students: Jan Eric Haßler, Kai Viktor Freissler
Commentary	This slide is about Real-Time Rendering Contest Results 3/3. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: Students: Jan Eric Haßler, Kai Viktor Freissler
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Real-Time Rendering Contest Results 3/3' into a causal sentence instead of repeating the slide title?
Page 6	1.1 Course Organization
Source cue	Details about the lecture course
Commentary	This slide is about 1.1 Course Organization. Treat it as relevance information. It tells you how the lecture, exercises, programming tasks, and exam preparation connect. The visible cue is: Details about the lecture course
Why it matters	Course logistics often reveal which topics are practiced, assessed, or expected in code.
Check yourself	Can you connect '1.1 Course Organization' to an exercise, project task, or exam-preparation action?

Page 7	Teachers
Source cue	- Lectures / timo.ropinski@uni-ulm.de / Office hours upon request / - Exercises / Poonam Aditya Sawant
Commentary	This slide is about Teachers. Treat it as relevance information. It tells you how the lecture, exercises, programming tasks, and exam preparation connect. The visible cue is: - Lectures / timo.ropinski@uni-ulm.de / Office hours upon request / - Exercises / Poonam Aditya Sawant
Why it matters	Course logistics often reveal which topics are practiced, assessed, or expected in code.
Check yourself	Can you connect 'Teachers' to an exercise, project task, or exam-preparation action?
Page 8	Course Format
Source cue	- Lectures / - Tuesday (14-16) and every other Wednesdays (12-14) / - Room O28/1002 / - Specific dates/modalities in Moodle (password: ICG26SUMMER) / - Lecture slides
Commentary	This slide is about Course Format. Treat it as relevance information. It tells you how the lecture, exercises, programming tasks, and exam preparation connect. The visible cue is: - Lectures / - Tuesday (14-16) and every other Wednesdays (12-14) / - Room O28/1002 / - Specific dates/modalities in Moodle (password: ICG26SUMMER) / - Lecture slides
Why it matters	Course logistics often reveal which topics are practiced, assessed, or expected in code.
Check yourself	Can you connect 'Course Format' to an exercise, project task, or exam-preparation action?
Page 9	Course Format
Source cue	- Exercises / - Every other Wednesday (12-14) / - Room O28/1002 / - Programming in C/C++ (introduction available upon request) / - Enroll in Moodle
Commentary	This slide is about Course Format. Treat it as relevance information. It tells you how the lecture, exercises, programming tasks, and exam preparation connect. The visible cue is: - Exercises / - Every other Wednesday (12-14) / - Room O28/1002 / - Programming in C/C++ (introduction available upon request) / - Enroll in Moodle
Why it matters	Course logistics often reveal which topics are practiced, assessed, or expected in code.
Check yourself	Can you connect 'Course Format' to an exercise, project task, or exam-preparation action?
Page 10	Exercises
Source cue	- Exercise / - Exercise 1 – Rendering Pipeline / - Exercise 2 – Primitive Types and Shaders / - Exercise 3 – Geometric Transformations / - Exercise 4 – Projection and Clipping

Commentary	This slide is about Exercises. Treat it as relevance information. It tells you how the lecture, exercises, programming tasks, and exam preparation connect. The visible cue is: - Exercise / - Exercise 1 – Rendering Pipeline / - Exercise 2 – Primitive Types and Shaders / - Exercise 3 – Geometric Transformations / - Exercise 4 – Projection and Clipping
Why it matters	Course logistics often reveal which topics are practiced, assessed, or expected in code.
Check yourself	Can you connect 'Exercises' to an exercise, project task, or exam-preparation action?

Page 11	Successful Course Completion
Source cue	- Pass the oral/written exam / - Dates to be announced / - „Notenbonus“ possible / - 50% success rate of exercise solutions for n-1 sheets / - Only running solutions are marked
Commentary	This slide is about Successful Course Completion. Treat it as relevance information. It tells you how the lecture, exercises, programming tasks, and exam preparation connect. The visible cue is: - Pass the oral/written exam / - Dates to be announced / - „Notenbonus“ possible / - 50% success rate of exercise solutions for n-1 sheets / - Only running solutions are marked
Why it matters	Course logistics often reveal which topics are practiced, assessed, or expected in code.
Check yourself	Can you connect 'Successful Course Completion' to an exercise, project task, or exam-preparation action?

Page 12	Course Overview
Source cue	Background / Journey along the / Rendering Pipeline / Lighting Effects / Extrahierte Tabellen:
Commentary	This slide is about Course Overview. Treat it as relevance information. It tells you how the lecture, exercises, programming tasks, and exam preparation connect. The visible cue is: Background / Journey along the / Rendering Pipeline / Lighting Effects / Extrahierte Tabellen:
Why it matters	Course logistics often reveal which topics are practiced, assessed, or expected in code.
Check yourself	Can you connect 'Course Overview' to an exercise, project task, or exam-preparation action?

Page 13	Literature
Source cue	- P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer Graphics / (4th Edition), AK Peters 2016. / - D. Shreiner, G. Sellers, J. Kessenich, B. Licea-Kane: OpenGL Programming / Guide: The Official Guide to Learning OpenGL / (8th Edition), Addison-Wesley 2013.

Commentary	This slide is about Literature. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer Graphics / (4th Edition), AK Peters 2016. / - D. Shreiner, G. Sellers, J. Kessenich, B. Licea-Kane: OpenGL Programming / Guide: The Official Guide to Learning OpenGL / (8th Edition), Addison-Wesley 2013.
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'Literature'?

Page 14	Relevant Chapters and Link to Lecture Course
Source cue	- For self study
Commentary	This slide is about Relevant Chapters and Link to Lecture Course. Treat it as relevance information. It tells you how the lecture, exercises, programming tasks, and exam preparation connect. The visible cue is: - For self study
Why it matters	Course logistics often reveal which topics are practiced, assessed, or expected in code.
Check yourself	Can you connect 'Relevant Chapters and Link to Lecture Course' to an exercise, project task, or exam-preparation action?

Page 15	(1.1 Course Organization)
Source cue	1.2 Computer Graphics Overview / 1.3 Pixel-based Representations / 1.4 3D Models / 1.5 Algorithmic Paradigms
Commentary	This slide is about (1.1 Course Organization). Treat it as relevance information. It tells you how the lecture, exercises, programming tasks, and exam preparation connect. The visible cue is: 1.2 Computer Graphics Overview / 1.3 Pixel-based Representations / 1.4 3D Models / 1.5 Algorithmic Paradigms
Why it matters	Course logistics often reveal which topics are practiced, assessed, or expected in code.
Check yourself	Can you connect '(1.1 Course Organization)' to an exercise, project task, or exam-preparation action?

Page 16	1.2 Computer Graphics Overview
Source cue	Relation of the field with respect to other disciplines
Commentary	This slide is about 1.2 Computer Graphics Overview. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: Relation of the field with respect to other disciplines
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn '1.2 Computer Graphics Overview' into a causal sentence instead of repeating the slide title?

Page 17	Computer Graphics History
Source cue	- Origins in art / - Perspective, color, etc. / - Computer graphics milestones [Reichardt, 1968] / - 1960 – „Computer Graphics“, William Fetter (Boeing) / - 1962 – Spacewars, Steve Russel (MIT) Computer
Commentary	This slide is about Computer Graphics History. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Origins in art / - Perspective, color, etc. / - Computer graphics milestones [Reichardt, 1968] / - 1960 – „Computer Graphics“, William Fetter (Boeing) / - 1962 – Spacewars, Steve Russel (MIT) Computer
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Computer Graphics History' changes positions before rasterization?

Page 18	Computer Graphics History
Source cue	Sutherland, Utah (1970) Wireframe
Commentary	This slide is about Computer Graphics History. Read it as boundary logic. Identify what is inside, what is outside, what can be trivially accepted/rejected, and where intersections are created. The visible cue is: Sutherland, Utah (1970) Wireframe
Why it matters	Clipping decides what geometry is allowed to reach rasterization and can create new boundary vertices.
Check yourself	Can you decide what is accepted, rejected, or newly intersected in 'Computer Graphics History'?

Page 19	Computer Graphics History
Source cue	Sutherland, Utah (1970) Turner Whitted, Siggraph (1979)
Commentary	This slide is about Computer Graphics History. Read it as boundary logic. Identify what is inside, what is outside, what can be trivially accepted/rejected, and where intersections are created. The visible cue is: Sutherland, Utah (1970) Turner Whitted, Siggraph (1979)
Why it matters	Clipping decides what geometry is allowed to reach rasterization and can create new boundary vertices.
Check yourself	Can you decide what is accepted, rejected, or newly intersected in 'Computer Graphics History'?

Page 20	Computer Graphics History
Source cue	The Abyss (1989) Terminator 2 (1991) Jurassic Park (1993)
Commentary	This slide is about Computer Graphics History. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The Abyss (1989) Terminator 2 (1991) Jurassic Park (1993)
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.

Check yourself	Can you turn 'Computer Graphics History' into a causal sentence instead of repeating the slide title?
Page 21	Computer Graphics History
Source cue	Ikea (2014)
Commentary	This slide is about Computer Graphics History. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: Ikea (2014)
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Computer Graphics History' into a causal sentence instead of repeating the slide title?
Page 22	Related Disciplines
Source cue	- Human-Computer Interaction / - Virtual and Augmented Reality / - Visualization / - Image Processing / - Computer Vision
Commentary	This slide is about Related Disciplines. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Human-Computer Interaction / - Virtual and Augmented Reality / - Visualization / - Image Processing / - Computer Vision
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Related Disciplines' into a causal sentence instead of repeating the slide title?
Page 23	Applications
Source cue	- Computer games / - Animation movies / - Special effects in movies / - CAD/CAM [CryEngine 5.7, CryTek 2022] / - Simulation
Commentary	This slide is about Applications. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Computer games / - Animation movies / - Special effects in movies / - CAD/CAM [CryEngine 5.7, CryTek 2022] / - Simulation
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Applications' into a causal sentence instead of repeating the slide title?
Page 24	1.3 Pixel-based Representations
Source cue	What is a pixel?
Commentary	This slide is about 1.3 Pixel-based Representations. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: What is a pixel?

Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn '1.3 Pixel-based Representations' into a causal sentence instead of repeating the slide title?
Page 25	Raster Images
Source cue	[„Bibliothek“ by press office Ulm University]
Commentary	This slide is about Raster Images. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: [„Bibliothek“ by press office Ulm University]
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Raster Images'?
Page 26	Pixel
Source cue	- Name derived from Word for Picture Element / - Pixel is a sample with associated sample values (=pixel components) / - Color / - Transparency (=alpha value) / - Depth (i.e., distance from viewer (=z value))
Commentary	This slide is about Pixel. Read it as an occlusion decision. Decide whether the method reasons about objects, image regions, rays, or per-fragment depth comparisons. The visible cue is: - Name derived from Word for Picture Element / - Pixel is a sample with associated sample values (=pixel components) / - Color / - Transparency (=alpha value) / - Depth (i.e., distance from viewer (=z value))
Why it matters	Visibility decides which generated candidates are actually seen from the current viewpoint.
Check yourself	Can you decide whether 'Pixel' works per object, per image region, per ray, or per fragment?
Page 27	Color
Source cue	- In real world color is given / by color spectrum Day Light Light Bulb / Fluorescent Tube White LED / - In computer graphics color is represented through color models / - RGB color model (additive)
Commentary	This slide is about Color. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - In real world color is given / by color spectrum Day Light Light Bulb / Fluorescent Tube White LED / - In computer graphics color is represented through color models / - RGB color model (additive)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Color'?
Page 28	LCD Raster Screen

Commentary	This slide is about 3D Model Types. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Models specify the structures to be displayed / - Three commonly used model types exist / - Implicit surfaces / - Polygonal models / - Volumetric models
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn '3D Model Types' into a causal sentence instead of repeating the slide title?

Page 32	Implicit Surfaces
Source cue	- Set of all points that fulfill $x, y, z = 0$ / - All points with $x, y, z < 0$ define solid bound by $x, y, z = 0$ / - Display possibilities / - Search for null points / - Approximate with polygons
Commentary	This slide is about Implicit Surfaces. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Set of all points that fulfill $x, y, z = 0$ / - All points with $x, y, z < 0$ define solid bound by $x, y, z = 0$ / - Display possibilities / - Search for null points / - Approximate with polygons
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Implicit Surfaces' into a causal sentence instead of repeating the slide title?

Page 33	Quadrics
Source cue	- Surfaces represented by quadric equations / Ellipsoid Single Hyperboloid Double Hyperboloid / $x^2 y^2 z^2$ / $x, y, z = ++-1$ $x, y, z = +- -1$ $x, y, z = +- +1$ / $x^2 y^2 z^2$
Commentary	This slide is about Quadrics. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Surfaces represented by quadric equations / Ellipsoid Single Hyperboloid Double Hyperboloid / $x^2 y^2 z^2$ / $x, y, z = ++-1$ $x, y, z = +- -1$ $x, y, z = +- +1$ / $x^2 y^2 z^2$
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Quadrics' into a causal sentence instead of repeating the slide title?

Page 34	Object Representation
Source cue	Point Cloud (3D Scan) Binary Data (MRT) Signed Distance Fields (SDF) Meshes

Commentary	This slide is about Object Representation. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: Point Cloud (3D Scan) Binary Data (MRT) Signed Distance Fields (SDF) Meshes
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Object Representation' into a causal sentence instead of repeating the slide title?

Page 35	Polygonal Models 1/3
Source cue	- Surface representation is approximated by polygons / - Polygons are made up of vertices and edges
Commentary	This slide is about Polygonal Models 1/3. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Surface representation is approximated by polygons / - Polygons are made up of vertices and edges
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Polygonal Models 1/3' into a causal sentence instead of repeating the slide title?

Page 36	Polygonal Models 1/4
Source cue	- Surface representation is approximated by polygons (mostly triangles/quads) / - Polygons are made up of vertices and edges / Valence / Degree Adjacent Edges 1-Ring Neighborhood
Commentary	This slide is about Polygonal Models 1/4. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Surface representation is approximated by polygons (mostly triangles/quads) / - Polygons are made up of vertices and edges / Valence / Degree Adjacent Edges 1-Ring Neighborhood
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Polygonal Models 1/4'?

Page 37	Polygonal Models 2/4
Source cue	- Surface representation are often approximated by triangular polygons / - Triangles consist of three vertices and three edges / - Display achieved through rendering by raster conversion of triangles
Commentary	This slide is about Polygonal Models 2/4. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Surface representation are often approximated by triangular polygons / - Triangles consist of three vertices and three edges / - Display achieved through rendering by raster conversion of triangles

Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Polygonal Models 2/4'?
Page 38	Polygonal Models 3/4
Source cue	- Various attributes are associated with vertices of a polygonal model / - Colors / - Material properties / - Textures / - ...
Commentary	This slide is about Polygonal Models 3/4. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Various attributes are associated with vertices of a polygonal model / - Colors / - Material properties / - Textures / - ...
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Polygonal Models 3/4'?
Page 39	Polygonal Models 4/4
Source cue	- Surface normal required in many cases / - Normal is calculated through cross product: / - Example use cases /  / 2 3 3 2
Commentary	This slide is about Polygonal Models 4/4. Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Surface normal required in many cases / - Normal is calculated through cross product: / - Example use cases /  / 2 3 3 2
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Polygonal Models 4/4'?
Page 40	Volumetric Models
Source cue	- Space filling object representation through voxels (volume elements) / - Display possibilities / - Rendering of extracted surfaces / - Direct rendering through specialized algorithms
Commentary	This slide is about Volumetric Models. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Space filling object representation through voxels (volume elements) / - Display possibilities / - Rendering of extracted surfaces / - Direct rendering through specialized algorithms
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Volumetric Models'?
Page 41	Model Generation

Source cue	- Polygonal models are generated through 3D scanning or 3D modeling / (e.g., LIDAR, Blender, Cinema 4D, Maya) / - Volumetric models are generated through imaging or simulation / (e.g., CT, MRI, PET, CFD, seismic surveys)
Commentary	This slide is about Model Generation. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Polygonal models are generated through 3D scanning or 3D modeling / (e.g., LIDAR, Blender, Cinema 4D, Maya) / - Volumetric models are generated through imaging or simulation / (e.g., CT, MRI, PET, CFD, seismic surveys)
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Model Generation' into a causal sentence instead of repeating the slide title?

Page 42	Coordinate Systems
Source cue	- Three main coordinate system exist / - Model coordinate system – model specified in own coordinate system / - World coordinate system – models arranged in common coordinate system / - Eye/camera coordinate system – model in relation to camera / - To transfer between coordinate systems,
Commentary	This slide is about Coordinate Systems. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Three main coordinate system exist / - Model coordinate system – model specified in own coordinate system / - World coordinate system – models arranged in common coordinate system / - Eye/camera coordinate system – model in relation to camera / - To transfer between coordinate systems,
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Coordinate Systems'?

Page 43	Rendering
Source cue	- Rendering is the process to generate raster images from 3D models / - During rendering several steps are necessary / - Coordinate system transformation / - Visibility determination / - Illumination
Commentary	This slide is about Rendering. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Rendering is the process to generate raster images from 3D models / - During rendering several steps are necessary / - Coordinate system transformation / - Visibility determination / - Illumination
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Rendering'?

Page 44	1.5 Algorithmic Paradigms
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Source cue	Frequently found algorithm strategies
Commentary	This slide is about 1.5 Algorithmic Paradigms. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: Frequently found algorithm strategies
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn '1.5 Algorithmic Paradigms' into a causal sentence instead of repeating the slide title?

Page 45	Photorealistic Rendering
Source cue	- Goal: simulate reality / [Embree Ray Tracing Website - http://embree.github.io/] [Maisch & Ropinski, Eurographics 2017] [Hendrik vann Jenssen]
Commentary	This slide is about Photorealistic Rendering. Read it as an occlusion decision. Decide whether the method reasons about objects, image regions, rays, or per-fragment depth comparisons. The visible cue is: - Goal: simulate reality / [Embree Ray Tracing Website - http://embree.github.io/] [Maisch & Ropinski, Eurographics 2017] [Hendrik vann Jenssen]
Why it matters	Visibility decides which generated candidates are actually seen from the current viewpoint.
Check yourself	Can you decide whether 'Photorealistic Rendering' works per object, per image region, per ray, or per fragment?

Page 46	Non-Photorealistic Rendering (NPR)
Source cue	- Goal: abstract reality / [Suggestive contour line drawings, DeCarlo 2003]
Commentary	This slide is about Non-Photorealistic Rendering (NPR). Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Goal: abstract reality / [Suggestive contour line drawings, DeCarlo 2003]
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Non-Photorealistic Rendering (NPR)'?

Page 47	Image-Based Algorithms
Source cue	- Number of computations direct proportional to number of pixels / for all Pixels do { / for all SceneObjects do { / - Benefits / - Easy implementation
Commentary	This slide is about Image-Based Algorithms. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Number of computations direct proportional to number of pixels / for all Pixels do { / for all SceneObjects do { / - Benefits / - Easy implementation
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.

Check yourself	Can you turn 'Image-Based Algorithms' into a causal sentence instead of repeating the slide title?
Page 48	Object-Based Algorithms
Source cue	- Number of computations direct proportional to number of 3D models / for all SceneObjects do { / for all Pixels do { / - Benefits / - Fast computation
Commentary	This slide is about Object-Based Algorithms. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - Number of computations direct proportional to number of 3D models / for all SceneObjects do { / for all Pixels do { / - Benefits / - Fast computation
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Object-Based Algorithms' into a causal sentence instead of repeating the slide title?
Page 49	are assigned
Source cue	- Raster images are formed of pixels to which colors, transparencies and depth / are assigned / polygonal models / - Most frequent building blocks of polygonal models are triangles, / which are formed of vertices and edges
Commentary	This slide is about are assigned. Read it as continuous-to-discrete conversion. The core question is which samples are covered and which interpolated values each fragment receives. The visible cue is: - Raster images are formed of pixels to which colors, transparencies and depth / are assigned / polygonal models / - Most frequent building blocks of polygonal models are triangles, / which are formed of vertices and edges
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'are assigned'?
Page 50	Literature and other sources used in this chapter
Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Commentary	This slide is about Literature and other sources used in this chapter. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Literature and other sources used in this chapter' into a causal sentence instead of repeating the slide title?
Page 51	Peters 2016. (Chapters 1, 3 & 21)

Source cue	- Text Books / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer Graphics (4th Edition), AK / Peters 2016. (Chapters 1, 3 & 21) / - Crassin, Cyril, Fabrice Neyret, Miguel Sainz, Simon Green, and Elmar Eisemann. "Interactive / indirect illumination using voxel cone tracing." In Computer Graphics Forum, vol. 30, no. 7, pp.
Commentary	This slide is about Peters 2016. (Chapters 1, 3 & 21). Read it as local shading. Identify normal, light direction, view direction, material coefficients, and where the computation is evaluated. The visible cue is: - Text Books / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer Graphics (4th Edition), AK / Peters 2016. (Chapters 1, 3 & 21) / - Crassin, Cyril, Fabrice Neyret, Miguel Sainz, Simon Green, and Elmar Eisemann. "Interactive / indirect illumination using voxel cone tracing." In Computer Graphics Forum, vol. 30, no. 7, pp.
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Peters 2016. (Chapters 1, 3 & 21)'?

01.1 Course Organization

Source location: Slides around 6-15

Commentary

The organizational slides matter because they explain how the theory and programming parts fit together. The exercises are not separate from the lecture; they are the practical version of the same pipeline ideas. When the course says that slides are generally not self-explanatory, that is a warning: the bullet points are anchors, not a textbook.

Treat every exercise sheet as an applied checkpoint. Rendering pipeline, primitive types, shaders, transformations, projection, clipping, and rasterization are introduced in the lectures and then turned into OpenGL tasks. If a topic appears both in a lecture and an exercise, it is more likely to be exam relevant.

Mental model

Lecture slides give the vocabulary; exercises force you to connect that vocabulary to code and visible output.

Check yourself

Can you list the five exercise themes and connect each one to a later lecture chapter?

01.2 Computer Graphics Overview

Source location: Section 1.2

Commentary

Computer graphics is about generating images from descriptions. The description can be pixel-based, object-based, physically motivated, artistic, or algorithmic. The common thread is that the computer must decide what color belongs at image samples.

Interactive graphics adds a time constraint. The image is not computed once and admired; it must respond to camera motion, user input, animation, and changing state. That is why real-time rendering often uses

approximations and specialized GPU stages.

Mental model

Graphics is controlled image generation. Interactive graphics is controlled image generation under a strict time budget.

Check yourself

What changes when a renderer must be interactive rather than offline?

01.3 Pixel-Based Representations

Source location: Section 1.3

Commentary

A pixel-based representation stores an image directly as samples on a grid. It is easy to display and edit locally, but it does not know the underlying 3D scene. If you zoom into a raster image, you reveal the sampling grid, not more geometry.

This is the first place where sampling becomes important. Resolution, color depth, aliasing, and frame time are not cosmetic details. They decide what information can be represented and how much data must be processed per second.

Mental model

A raster image is already the answer. It contains color samples, not the geometric reasons behind them.

Check yourself

Why can a pixel image be easy to display but hard to reinterpret as a 3D scene?

01.4 3D Models

Source location: Section 1.4

Commentary

A 3D model is a structured description of objects before they become pixels. It may contain vertices, primitives, topology, normals, materials, textures, and transformations. Unlike a pixel image, it can be viewed from different camera positions.

The course mostly follows object-based rendering: start from model data, transform it, project it, rasterize it, shade it, test visibility, and finally update pixels. Later chapters explain these steps one by one.

Mental model

A model is not an image. It is a cause from which many possible images can be generated.

Check yourself

Which model attributes can influence final color besides vertex positions?

01.5 Algorithmic Paradigms

Commentary

The lecture contrasts ways to generate images. Rasterization works from objects toward pixels and is the dominant real-time pipeline. Ray-based methods work from image samples into the scene and are powerful for visibility and lighting but can be more expensive.

Do not memorize these paradigms as names only. Ask what direction information flows. Does the algorithm start from geometry and find covered pixels, or from pixels/rays and find visible geometry? That direction explains the strengths and weaknesses.

Mental model

Rasterization asks, 'Which samples does this primitive cover?' Ray casting asks, 'What does this sample see?'

Check yourself

Which paradigm fits OpenGL's standard real-time pipeline most directly?

End-of-lecture summary

If you remember only one thing from Lecture 01, remember this: This lecture is the orientation layer for the whole course. It explains why interactive computer graphics is a pipeline problem: a scene must be represented, processed under time constraints, sampled into an image, and displayed fast enough for interaction.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.